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FREEDOM-TO-OPERATE OPINION

Helios FMCW LiDAR — Field of Use: Industrial / Robotics

Private & privileged | Prepared for Cobalt Photonics Limited | 3 December 2025

1. Instructions

We have been instructed by Cobalt Photonics Limited to provide a freedom-to-operate opinion in respect of the Helios coherent-detection LiDAR architecture, in connection with a proposed licence into the industrial and robotics field of use. In particular we were asked to assess the risk posed by third-party patent US10,114,202, identified during clearance searching.

2. The cited patent (US10,114,202)

US10,114,202 (“the Cited Patent”), held by a US sensor manufacturer, claims an FMCW detection system. Independent claim 1 requires, among other elements, a “mechanically rotating optical scanner” coupled to the detection chain. The dependent claims add features consistent with that mechanical-scanning architecture.

3. Analysis

The Helios architecture performs beam-steering using a solid-state on-chip optical phased array; it contains no mechanically rotating optical scanner. On a proper construction of claim 1, the “mechanically rotating optical scanner” limitation is essential and is not present, expressly or by equivalents, in the Helios architecture. We have reviewed the prosecution history of the Cited Patent, which confirms the applicant added the mechanical-scanner limitation to overcome prior art, narrowing the claim and supporting a prosecution-history-estoppel argument against any doctrine-of-equivalents reading.

4. Opinion

In our opinion, the manufacture, use and sale of products implementing the Helios architecture within the industrial and robotics field of use would not infringe any valid claim of US10,114,202. The principal distinguishing feature is the solid-state on-chip beam-steering used by Helios in place of the mechanically rotating scanner required by the Cited Patent’s claims. We consider the risk of an infringement finding to be low.

5. Qualifications

This opinion is based on the patents and information available to us as at the date of this opinion and is limited to the Cited Patent. It does not constitute a guarantee of non-infringement and is subject to the usual qualifications. It is provided to Cobalt Photonics Limited and may be disclosed to Tessandri Instruments S.p.A. and its advisers in connection with Project Indigo on a non-reliance basis.

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